Department of Mechanical Engineering

ME3 Design, Make and Test

DMT Group 5C

Rocket Body & Feed System

Product Presentation & Test Report

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Date: 3rd June 2025

Document History:

Version	Date	Description
1.0	01-06-2025	First draft
2.0	03-06-2025	Finalised wording

Executive Summary

The super project aimed to deliver a fully integrated rocket, with this subassembly handling the storage and delivery of the ethanol and nitrous oxide propellants to the engine. The proposed solution employed a single tank divided by a free-moving piston, with the nitrous oxide self-pressurising to 50 bar at ambient temperature. All plumbing and structural components were encapsulated within a uniform diameter enabling compact integration with the body. This mass-optimised assembly was designed for reusability across 10 hot fires without disassembly and incorporated a piping spacer that also supported the TVC team's gimbal and actuators.

Hydrostatic testing proved that the designed structural and sealing integrities were sufficient to withstand 1.6x the operating pressure. Cold flow and hot fire testing saw the system successfully deliver propellant to sustain a 10 second burn time. Measured mass flow rates in the fuel line achieved 78% and 70% of the target: 0.124 and 0.112 kg/s for the first and second hot fires respectively. The rocket remained fully intact with no component damage, highlighting the effectiveness of the design for reusability. Friction-related pressure drops were negligible and remained below predicted values. This was true for the cold flow and both hot fire tests and demonstrated the piston's consistently smooth motion throughout.

To gain a better understanding of the system's behaviour, improvements in data acquisition are essential. The current sensor setup limited the ability to capture the dynamic interactions between subsystems, which influence each other through feedback loops. Adding more pressure transmitters at key interfaces or incorporating flow meters in the propellant lines would improve data for later analysis.

The successful integration of all subassemblies enabled a major milestone, marking this project as **the first ever vertical student-led hot fire test ever performed in the UK**.

Overview of the Super-project

This super-project designed, made, and tested a reusable liquid bi-propellant rocket propulsion and flight system in collaboration with Imperial College London Rocketry (ICLR), with a target impulse of 10 kNs for 10 possible hot fires. The system incorporates regenerative cooling, thrust vector control via in-house manufactured actuators, and an integrated fuel and body system, building upon the work of the previous 2023-24 DMT project. The product was validated with vertical cold flow and hot fire testing at Silwood Park, **marking the first ever vertical static hot fire by a student group in the UK**. The project, overseen by Peter Johnson, is split into three sub-assemblies:

1. The Thrust Vector Control (TVC) group developed a universal joint gimballing system and in-house manufactured electro-mechanical actuators for thrust vectoring of the engine. This system allows the engine to be steered within a range of motion allowing for controlled positioning of the engine and the rocket.
2. The Engine group developed the UK's first stainless steel regeneratively cooled rocket engine. Incorporating regenerative cooling via additive manufacturing, the engine produces thrust by injecting, combusting and accelerating the propellants, whilst achieving sufficiently low steady-state temperatures of operation.
3. The Body and Feed group developed a fully integrated rocket body and feed system. Using a single self-pressurising piston tank design and a piping system contained within the same silhouette, the ethanol and nitrous oxide propellants are fed through the TVC stage to the engine intake at sufficient pressure and mass flow rate to sustain the target rocket burn duration.

The full super-project assembly is shown in Figure 1.

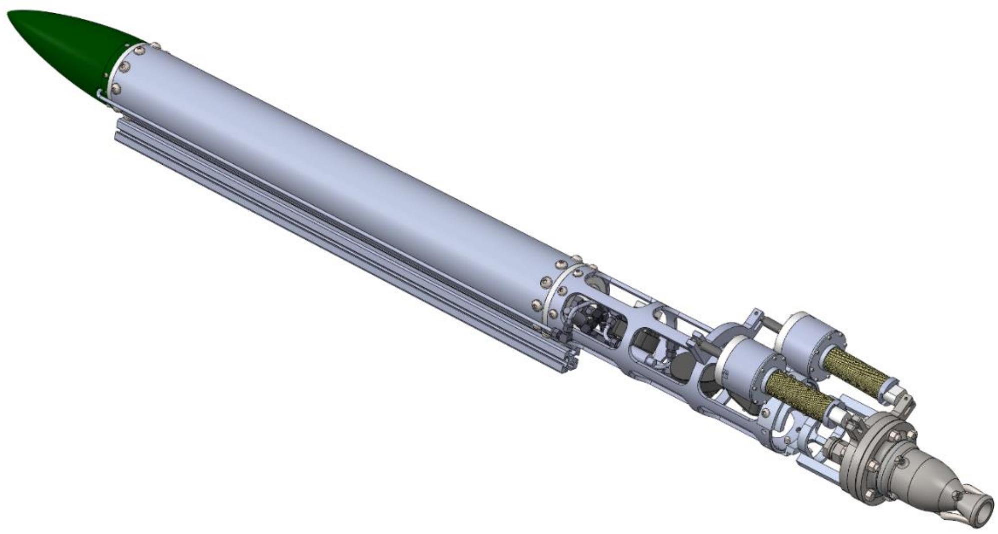

Figure 1: Full super-project assembly, incorporating Thrust Vector Control, Engine and Fuel and Body group subassemblies

Project Description

Background & Introduction

Rocketry has traditionally relied on tank pressurisation using a simple blow down method, where inert gas is introduced into the propellant tanks. While this approach is straightforward, it results in significant mass overhead due to the need for separate high-pressure vessels and ullage gas. Since the 1960s, the aerospace industry has employed positive expulsion feed systems, such as pistons, bladders and diaphragms, to ensure propellant delivery in zero-gravity environments (1).

Positive expulsion systems keep a constant pressure with a piston as physical separation between fuel and oxidiser. This concept gained popularity in amateur and experimental rocketry towards the late 1990s with the advent of nitrous oxide-based propulsion, due to its self-pressurisation properties (2). This not only removes the need for external pressurisation systems but also simplifies propulsion as it requires no active control once initiated.

Nitrous oxide (nitrous) is only just subcritical at ambient temperature (20°C). In this state, small pressure changes across an emptying tank result in immediate production of extra vapour; the gas works to maintain a tank pressure of around 50 bar (3). In contrast, conventional blow-down systems lose tank pressure at a much higher rate, reducing the thrust to mass ratio across the duration of the fire.

There is a history of student teams achieving rockets of several kilonewtons worth of thrust with similar designs. The Polish student team AGH Space Systems developed the *Turbulence* rocket, rated at 4 kN, in 2018, which used a single integrated diaphragm tank for N_2O and ethanol (4). By the early 2020s groups like Half Cat Rocketry openly published designs such as the *Mojave Sphynx* (5) and detailed guides on propellant driven piston pressurization, emphasising its advantages (6). Compared to bladder-based feed systems, piston tanks are more space-efficient, easier to manufacture and less prone to deformation or flow inconsistencies (7). Last year's DMT successfully implemented a piston tank, relying on a heavier tie-rod frame.

These precedents demonstrate the feasibility of piston-based feed systems in amateur rocketry, while also highlighting the engineering challenges involved. Among them is the need for effective sealing for the piston and bulkheads, as well as mass optimisation and structural integration. Informed by past projects and research, our design aimed to develop a compact, piston-driven feed system for a nitrous oxide and ethanol bipropellant engine. The system was designed to deliver 1kN of thrust over 10 seconds at target mass flow rates of 0.16 kg/s and 0.37 kg/s of fuel and oxidiser respectively, and a post-injector chamber pressure of approximately 30 bar. The PDS also highlights the importance of weight optimisation, establishing a 15kg dry mass target while maintaining sealing integrity and withstanding pressures exceeding operating conditions. The tank was also required to be reusable for 10 hot fires without the need for disassembly.

Manufactured Subassembly Product

Final Assembled Product

The final subassembly, shown in

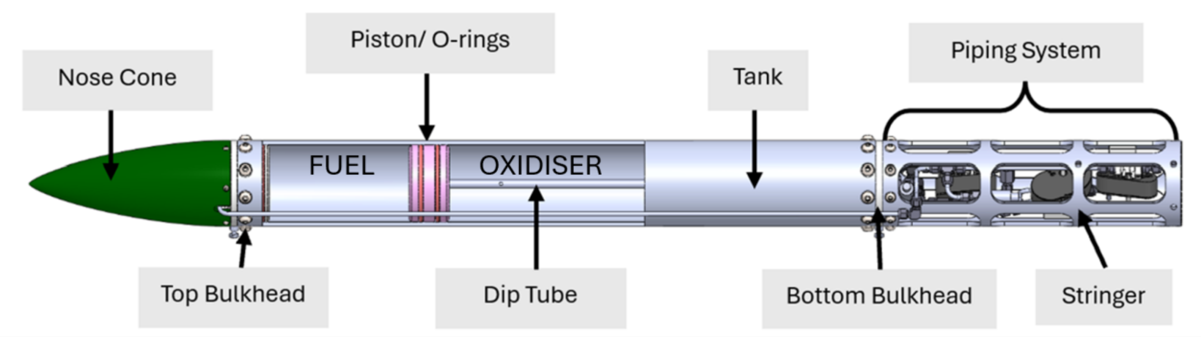

Figure 2, features a single tank pressure vessel that contains both fuel and oxidiser, separated by a piston. Nitrous was selected as the oxidiser due to its favourable thermodynamic properties, while ethanol is used as the fuel for its wide availability. The use of a single tank design simplified the manufacturing process and reduced the number of components, helping to keep the project within budget while optimising the weight of our subassembly.

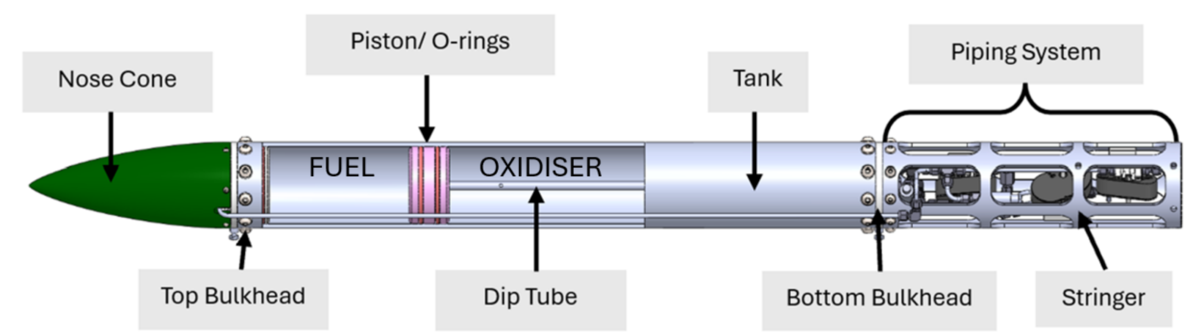

Figure 2 Full subassembly

Tank

The tank is made of Aluminium 6082-T6 alloy round tube. Aluminium alloys are the most popular choice for amateur liquid propellant tanks due to their high strength to weight ratio (9), ease of machining and wide availability in appropriate forms. The tube has an ID of 114.30 mm, thickness of 6.35 mm and a length of 959 mm. These dimensions were selected to hold the required propellent volumes for a 10 second burn time. The material has a density of 2700 kg/m^3 and a yield stress of 260 MPa, critical to ensure it can withstand high internal pressure. The calculated hoop stress of the tank was 47.6 MPa at 50 bar, giving a safety factor of 5.5. During manufacturing, it was discovered that the tank was too big to secure to match drill radial bolt holes with the bulkheads. Instead, a 3D-printed stencil was used to mark the hole locations. This proved more effective than manual measurement but still led to minor misalignments between the parts.

Piston & O-rings

The piston separates the nitrous and ethanol in the tank and is used to transfer pressure to the ethanol. The O-rings and their material selection are critical, as demonstrated by the infamous Challenger accident in January 1986, caused by O-ring failure. Our choice of Viton is based on literature detailing its properties (10) in propellant sealing applications under extremely harsh conditions. It provides excellent chemical compatibility and thermal resistance, critical for maintaining a pressure-tight seal throughout testing. The dimensions of the O-rings and their grooves were carefully designed to balance effective sealing with minimal friction and follows ISO 3601 standards. Two 5.33 mm cross-sectional diameter O-rings were selected. EN1A mild steel was the material selected for the piston as it is compatible with both ethanol and nitrous oxide, resisting corrosion in both environments (11). Using mild steel sliding within an aluminium cylinder also avoids issues such as cold welding or seizure common with aluminium-on-aluminium interfaces. The piston was turned in the STW and material was removed from its core for mass optimisation, in line with the PDS weight target.

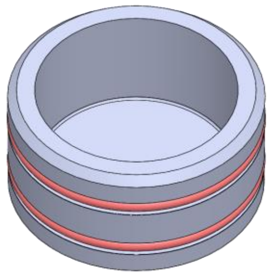

Figure 3 Piston with O-rings

Bulkheads

The bulkheads are made of aluminium 6082-T6 alloy and are designed to seal the tank to prevent leaks while withstanding higher-than-operating pressures. To remain within the maximum dry system mass of 15 kg specified in the PDS, the bulkheads were designed to be hollow. One of the primary issues encountered as a result was the limited accessibility for spanner tools when installing Swagelok fittings onto the bulkheads. Due to the constrained space, adjustments to the oxidiser line fittings required the removal of the dip tube, unnecessarily prolonging both assembly and testing processes. Two Viton O-rings of the same dimensions as ones used on the piston were selected for sealing. The bulkheads were CNC turned in the STW, where additional manually machined features including launch lugs and tank bolt holes were also added. Finally, the nose cone was secured to the top bulkhead via bolted connections, completing the tank subassembly.

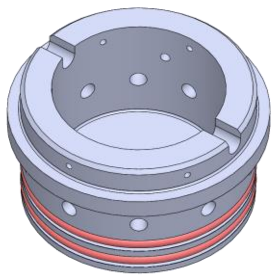

Figure 4 Top bulkhead with O-rings

Stringer

The 6082-T6 aluminium alloy stringer was outsourced for manufacturing to take advantage of the laser cutting process' increased precision. This material's high strength-to-weight ratio provided the necessary structural integrity while aiding mass optimisation. Cut-outs were incorporated into the design to ensure access for spanners and hands to tighten fittings during assembly. The stringer served as both the casing for the piping system and the interface for attaching the Thrust Vector Control (TVC) gimbal and actuator mechanisms. Holes drilled in the bottom section allowed for their gimbal structure to be bolted on with additional holes further up for actuator attachment. Two stringers were purchased for a small cost increase to account for the unknown testing orientation at the time.

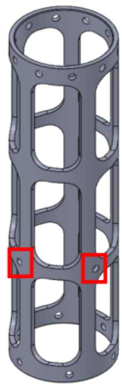

Figure 5 Stringer with actuator attachment highlighted

Plumbing & Dip Tube

The plumbing system consists of pipes, fittings, pressure transmitters (PTs) and valves. It includes three main lines: the fuel line, the oxidiser line and the safety vent line. The system was designed to be fully compatible with ICLR's servo valves and PTs. The Plumbing and Instrumentation Diagram (P&ID), shown in Figure 6, illustrates the required fittings as well as the placement of PTs and servo valves. Three PTs are integrated into the system: PT1 at the fuel tank, PT2 at the inlet to the fuel servo and PT3 at the oxidiser tank.

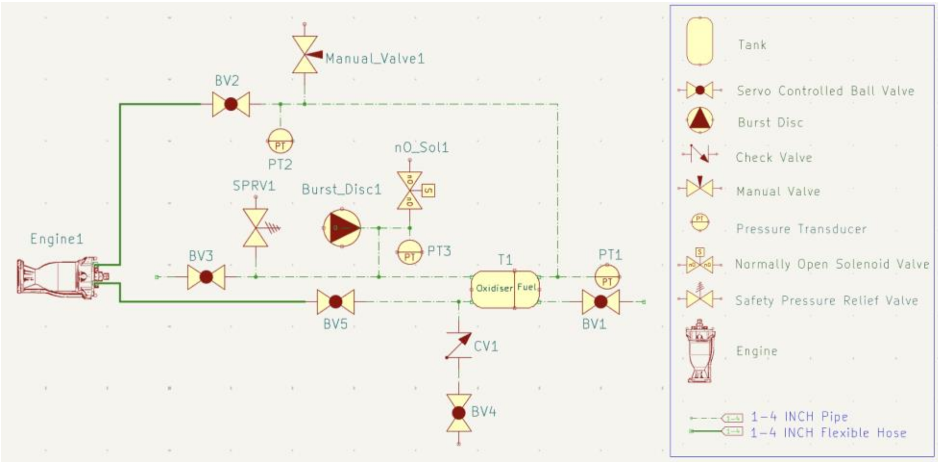

Figure 6 Full P&ID diagram

Two safety mechanisms were incorporated into the vent line. The first is a Safety Pressure Relief Valve (SPRV), which is a reusable component designed to progressively open at 55 bar. A single use burst disc is installed as a secondary safeguard in the event of SPRV failure. This component is designed to rupture at 70 ± 7 bar, to vent the pressurising oxidiser side of the piston. Together, these systems ensure safe operation in static fire testing.

During the filling process, the dip tube, visible in Figure 2, plays a dual role. First, acting as a stopper to indicate the bottom of the piston motion and ensure the correct ratio of fuel and oxidiser volume. Second, indicating when the appropriate level of liquid nitrous is filled, allowing it to vent out as a thick plume upon reaching two holes drilled 20% from its top end.

Operation of product

Before the filling process began, the piston was pushed to the end of its motion by pressurising the fuel tank with inert nitrogen gas until it reached the dip tube. To monitor the piston's position, the solenoid valve on the nitrous vent line was left open for the sound of escaping air to indicate the piston's movement down the tank. Once the sound stopped, it was assumed that the piston had come to rest against the dip tube. However, this is a somewhat unreliable indicator as the piston is not necessarily at the dip tube when it stops moving and could be jammed elsewhere.

The filling procedure begins by loading ethanol into the tank using a hose and pump, through Manual Valve 1 (Figure 6). Ethanol is pumped in until it begins to exit through Ball Valve 1 (BV1), indicating the tank is full. Any excess ethanol is collected in a container and disposed of appropriately. Once filling is complete, BV1 and the manual valves are closed. Nitrous is then filled remotely via an external servo through the designated check valve, which is connected to a pressurised nitrous oxide cylinder.

Budget Review

All incurred costs fell within the manufacturing phase, with a final total of £1,993.38, just below the revised £2,000 cap. The initial project budget was of £1,500, but an increase was approved to accommodate rising material and component costs, particularly in the main body and outsourcing of the stringer. Figure 7 shows the evolution of our budget breakdown, *(a)* before procurement began and *(b)* as the final expenditure. The former heavily favoured plumbing components, which accounted for nearly 60% of projected spending. As procurement progressed, costs increased across almost all categories, with the main body seeing the largest rise due to changes to the outsourced part. A detailed procurement list is provided in Appendix A. Testing equipment was the only area where spending decreased relative to the initial estimate, as several components and tools were borrowed from ICLR, including Allen and Torx keys, spanners, servo valves, a check valve, pressure transmitters and a solenoid valve.

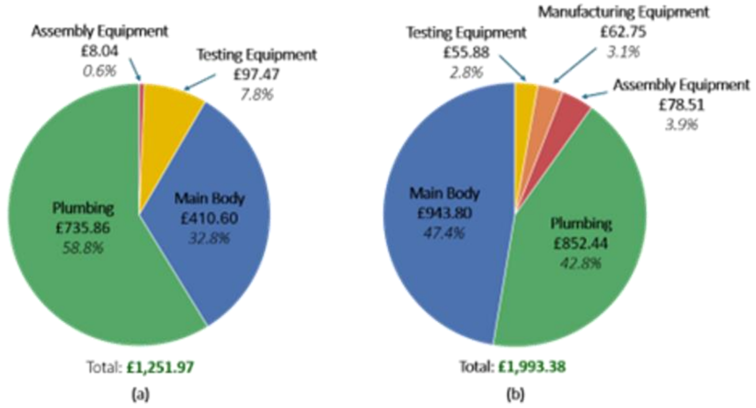

Figure 7 Budget Breakdown (a) at manufacturing gateway stage, (b) at testing stage

Outlook for the Developed Product

As it stands, the super-project's final product can be seen as a low-cost, reusable static hot fire and cold flow testing system. It has strong potential for an educational environment and, with some further development, of becoming a flight-capable rocket. Its modularity, reusability and relatively low-cost static testing makes it a promising tool for small scale propulsion experiments and training of future engineers.

A key short-term opportunity lies in educational implementation. A particularly relevant proposal is to adopt the system as the new first-year Mechanical Engineering lab at Imperial College London. Students would observe live cold flow and hot fire demonstrations before completing data analysis tasks. These would reinforce and demonstrate core thermo-fluids and stress analysis concepts including first law analysis for open systems, pressure drop and flow work, the steady flow energy equation (SFEE) and quasi-equilibrium processes within control volumes as well as strain analysis on cylindrical hollow bodies. The lab would offer a more dynamic and exciting introduction to engineering practice while providing exposure to the kind of work carried out in later years of the degree and in student-led initiatives such as Imperial College London Rocketry.

In parallel to its educational value, the final product has the potential to be further developed into a flight-ready rocket. As this design has been approached with realistic propulsion and structural constraints, achieving this goal would not require much rework, only targeted adaptations. Through collaboration with ICLR, who have already expressed interest in this direction, we have the possibility of launching the rocket at competitions such as the European Rocketry Challenge.

From an intellectual property standpoint, some elements of the system and subassemblies can be considered. While the piston and tank design themselves are not sufficiently novel for patenting, some integration features may be. For example, the use of a stringer doubling as both a piping spacer and mounting point for thrust vectoring system is a compact and efficient interfacing solution. Copyright protection would be applicable to the CAD models, and internal know-how (test protocols, tuning methods, etc) may gain relevance if the product moves towards commercialisation in the future.

However, at this stage, patenting is not recommended. Beyond novelty and innovation, the design is still early-stage, lacks a clear commercial route and there is no available budget to defend the IP. A more suitable strategy would be to publish or open-source the results. We propose releasing design files and performance data on a public GitHub repository to promote reproducibility, enable academic adoption and encourage student or amateur-led iteration. This would also secure our freedom to operate, ensuring the work cannot be appropriated by future IP claims.

In the medium to long term, a speculative yet realistic evolution of the system would be its development into a commercial teaching kit. This would involve combining the rocket with appropriate sensors and a data acquisition system into a modular, documented platform suitable for university laboratories or other technical teaching environments. A trial within Imperial's own teaching curriculum would serve as a valuable proof of concept.

Beyond this, market introduction would require further refinement in standardisation, manufacturing optimisation alongside improved safety and user experience. As the current system draws from multiple suppliers, consolidating sourcing, potentially outsourcing manufacturing, would reduce lead times and manufacturing overheads while improving consistency and standardised quality. The design should also be revisited with ease of manufacture and assembly in mind, for example improving ease of access to various piping fittings. Finally, clear documentation would be key for safe and repeatable operation, especially in educational settings where the system may be handled by non-specialists.

Hydrostatic Pressure and Leak Testing

Test Description & Justification

Before static hot fire testing, it is necessary to verify that each propellant tank can withstand pressures beyond its design point and remain leak tight. This is particularly critical in our piston-based design, where the only separation between fuel and oxidiser is the two piston O-rings. A failed seal could result in pre-mixing, severely compromising combustion control in the engine and posing a major explosion hazard. Explicit validation prior to propellant loading is therefore a prerequisite for safe operation.

This test was designed to assess the sealing performance at the tank's critical points, the two bulkheads and the piston, while also exposing the structure to over-pressurisation conditions to verify and compare predicted stresses against real measurements. A successful test was defined as one in which no visible leaks occurred at any O-ring interface, no permanent tank deformation was observed, and the total pressure drop across a one-hour period remained below 6 bar (i.e. 1 bar every 10 minutes). This threshold was deemed acceptable given the relatively short pressurisation duration (<10 minutes) before a hot fire. Following the *Pressure Equipment (Safety) Regulations 2016* (12), it is typical for pressure vessels in the UK to be tested to 1.43x their maximum allowable pressure. Testing was carried out to 80 bar (1.6x operating) to exceed the theoretical maximum triggering pressures of the 55 bar SPRV and 70 ± 7 bar burst disk. This verifies the tank would be the last of these components to fail.

Method

The two test setups, shown in Figure 9, are identical in assembly and procedure except for the top-end sealing method: first using the top bulkhead O-rings, then the piston O-rings. This order was chosen to avoid removing the more critical piston seals after testing, thereby minimising the risk of pre-mixing in subsequent hot firing. The top bulkhead seal integrity is revalidated during cold flow testing, so its removal posed no issue.

To validate theoretical predictions, the strain around the tank's radial holes (acting as stress concentrations) was measured under an 80-bar internal pressure. A stochastic pattern was applied to this region for Digital Image Correlation (DIC) to capture strain distribution and enable calculation of both axial and hoop stresses. As shown in Figure 8, the curved surface between holes was assumed sufficiently flat for 2D analysis. An image was taken every 5 seconds with a 4K camera, and a simple 10mm scale was marked on the pattern as a backup in case the applied stochastic pattern proved insufficient.

Figure 8 Applied DIC pattern with intended measurement region and backup 10mm markings shown

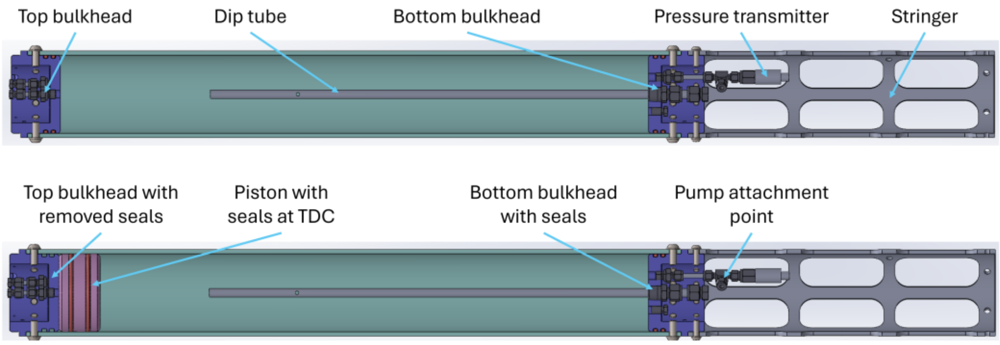

Figure 9 Hydrostatic tank assemblies

Once assembly was complete and the strain pattern applied, the tank was filled with water by standing it on its top end and opening two ports in the bottom bulkhead. A funnel was used to pour water in while allowing air to escape until both fittings overflowed, indicating absence of trapped air. This was a critical step, as compressible gases store significantly more explosive energy than incompressible liquids increasing the severity of any potential failure during pressurisation.

The stringer was then bolted to the bottom bulkhead to provide a fixed reaction point for expansion, allowing full extension to be captured by the camera positioned at the opposite end. The rig was relocated to the engine room, where pressurisation could be conducted safely behind blast doors (Figure 10), with a flexible hose fed through a small gap. Paper towels were placed beneath each end to make leaks visible from outside. The camera and pressure transmitter data logging was started simultaneously to synchronise timestamps, and the door was closed near fully before pressurising.

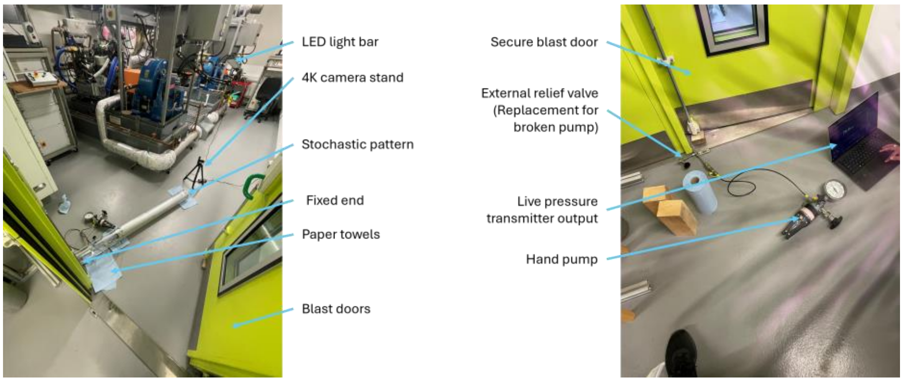

Figure 10 Hydrostatic testing setup inside (left) and outside (right) of the blast doors

Pressurisation was paused at 10 bar intervals to check for pressure stability as an indicator of leaks. As 80 bar was approached, pumping was slowed using the hand pump's fine adjustment mode to avoid overshooting. Upon reaching target pressure, the pump was placed inside the engine room and the system left pressurised for an hour.

After depressurisation via the external relief valve (Figure 10, right), the O-ring sealing points and Swagelok fittings were inspected for leaks. Staining of the blue paper towels at the less visible top end of the tank was used to identify any unnoticed leakage during the test.

Limitations & Sources of Error

There were some significant limitations which arose during completion of this test. Most notably, the pressure relief valve on the hand pump was broken and did not work to depressurise the system. To address this, a substitute valve was installed at the tank-side end of the hose. However, this reduced the effective reach of the hose and prevented the tank from being positioned far enough inside the blast doors to avoid interaction. As a result, sealing the room without disturbing the camera alignment or test setup became challenging, and the lighting conditions on the stochastic pattern were suboptimal. This also made real-time observation of the top bulkhead more difficult than planned.

The pressure sensor was initially calibrated to ambient lab conditions, where nearby experiments may have introduced a zero-error offset. The sensor was not recalibrated between the two test runs so both would operate on a comparable scale. Additionally, high frequency noise was present in recorded data, requiring application of moving average filtering. While this helped smooth the signal, an improperly chosen window size could have attenuated true signal features.

Strain measurement quality was limited by issues in both pattern application and camera setup. The black spray paint used was running low, resulting in an uneven and coarse speckle distribution.

Additionally, the camera lens' physical focus could only resolve a small central portion of the curved tank surface, leaving outer regions somewhat blurred. Repeated assembly and disassembly also led to progressive damage to the pattern around the bolts, with the primary region of interest nearly worn away by the final run. Only one end of the tank was monitored during testing, where the top bulkhead showed greater than expected settling movement during pressurisation. This suggests the bottom bulkhead introduced a similar additional compliance which would interfere with the strain measurements and warrants further inspection in future tests.

Predicted Performance

Stress analysis was conducted during design to evaluate global tank loading and local concentrations at the radial bolt holes. A force balance of the bulkheads, considering maximum and minimum wall areas (Appendix B), gave predicted axial stresses of 34.1 MPa and 46.3 MPa under an 80 bar internal pressure. Applying a stress concentration factor of 3 at the drilled holes yielded a local stress of 102.3 MPa.

Using a Young's modulus of 71 GPa for aluminium 6082-T6 (9), these stresses convert to tank strains of 0.00048 and 0.00065, or 0.46 mm to 0.62 mm extension across the full length. The local strain around the bolt holes is predicted to be 0.00144, equivalent to a 0.02 mm extension across the 1 mm hole length, which is used for direct DIC comparison.

Measured Performance & Discussion

Pressure data from both test runs was analysed in MATLAB. As shown in Figure 11 (top), pressure readings were aligned such that the 80 bar peaks coincide at zero on the time axis. Figure 11 (bottom) shows the corresponding pressure change rate in bar/min. Raw data was sampled at 50 Hz.

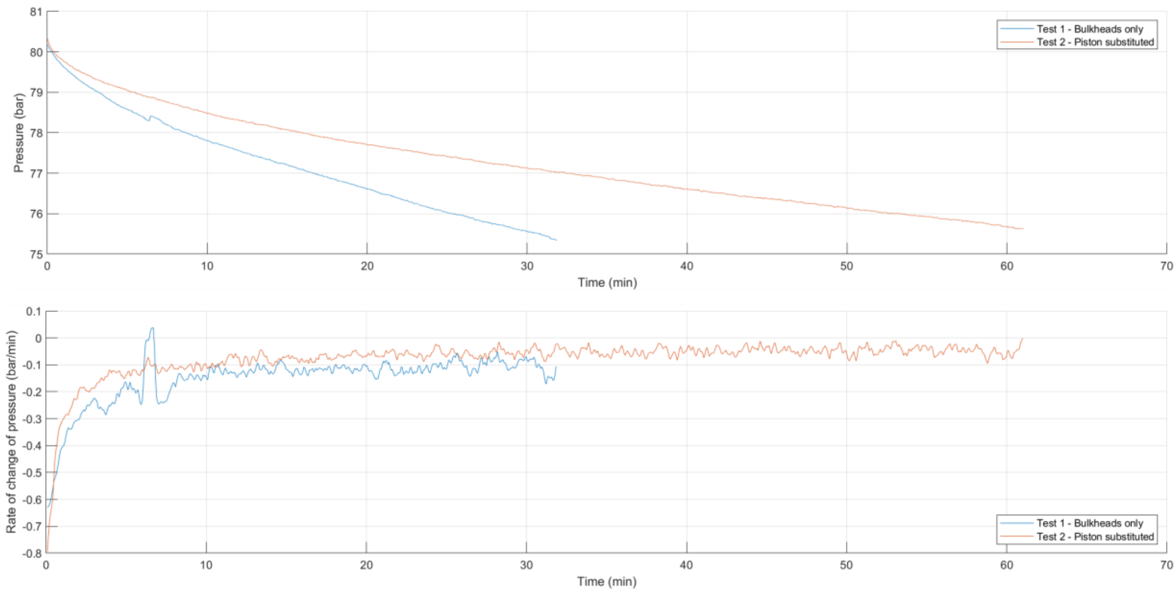

Figure 11 Smoothed bulkhead and piston hydrostatic test pressure readings against time (top) and leak rate (bottom)

A moving average (200-point window, 4 seconds) was applied to remove high-frequency noise. The data was then down sampled (1:200) and differentiated to obtain the pressure change rate. The first test was stopped after 30 minutes due to a considerable pressure drop, later traced to an under-tightened Swagelok fitting connecting the pump to the tank. Despite this, no O-ring leaks were observed, and the tank remained structurally intact at 80 bar, so the test was considered a success. For the second run, the leaking fitting was retightened but continued to leak at -0.05 bar/min, half of the -0.1 bar/min of the previous test. This suggested improper swaging rather than simple under-tightening, and the fitting was replaced in the final build. Total pressure loss during the second test was 4.3 bar over 60 minutes (0.72 bar every 10 minutes), within acceptable bounds (<6 bar/h).

Both tests exhibited a logarithmic decrease in rate of change of pressure, consistent with expectations for a leaking seal where flow rate depends on pressure differential. A transient spike at around 7 minutes in the bulkhead run was caused by accidental pump movement during repositioning inside the blast room. Interestingly, decay curves from both test runs tend to flatten around 77.4 bar, as detailed in Appendix C. This may indicate that the leak path partially sealed at this pressure or that components, potentially within the hand pump, settled during the first period of the test. Further testing across different starting pressures and fitting tightness is recommended to confirm this behaviour.

Strain DIC analysis was performed using ZEISS INSPECT Correlate and GIMP software. However, poor pattern quality prevented reliable local data extraction from ZEISS due to coarse spray application and damage from repeated reassembly necessitating pixel-based analysis on the 10 mm backup markings. As detailed in Appendix D, ZEISS output a 0.921 mm total extension across the tank length with pixel-based analysis giving a 0.873 mm extension which are approximately 50% greater than predicted. Pixel-based analysis also yielded a local extension of 0.0172 mm per 10 mm marking (0.00172 as a strain) which is 19% greater than predicted.

Despite the limited image quality, results suggest that DIC would be viable if patterning, lighting and focus were improved. However, measured strains approach the resolution limit of the camera: the smallest difference observed was just one pixel, below which any strains would be unresolvable. A bonded strain gauge would provide more accurate data at this fine scale and be less sensitive to tank shifting or bulkhead movement, both of which introduce an unknown degree of error into the current strain results.

Conclusion & Design Implications

This test successfully demonstrated that the tank could withstand the full 80 bar internal pressure without rupture or leakage, including across bulkheads and, more importantly, the piston seals. The only observed leak was traced to a damaged Swagelok fitting, which was replaced ahead of further testing. These outcomes are a key step before a safe full-system hot fire can be completed.

Strain measurements suggested a total tank extension of ~0.9 mm, on the same order of magnitude as the 0.6 mm theoretical prediction. However, limitations such as the mobile 'fixed' stringer end and poor DIC pattern quality reduce confidence in measured data. Bonded strain gauges could circumvent these issues by attaching directly to the region of interest.

The test also highlighted several design-for-testability issues. Bulkhead recesses, while effective for mass optimisation, complicated assembly and limited tool accessibility. Future designs should verify that test fittings can be installed or swapped using standard tools instead of requiring specific socket wrenches. Finally, the broken hand pump should be replaced, or a longer hose purchased, so the test setup can be positioned deeper within the engine room to avoid interference with the blast doors.

Static Fire Testing

Test description & Justification

For the project's supergroup assembly test, a static fire test was conducted. This included a cold flow and hot fire test which were both conducted at ICLR's test site at Silwood Park on the 24th of May 2025.

A cold flow test is a non-combustive procedure used to verify the rocket's feed system, using inert fluids in place of the actual propellants. Nitrogen, widely available and inexpensive, replaced the oxidiser (nitrous) while distilled water was used to simulate the fuel (ethanol). The cold flow test aimed to detect leaks, verify valve operation and PT functionality, confirming proper delivery of propellants to the engine. It also validated servo-valve actuation and timing. Additionally, the test enabled safe evaluation of ICLR's remote ground control system and data acquisition setup.

Figure 12 A frame from successful hot fire 2

Following a successful cold flow, the real propellants could be loaded, and a hot fire performed. The procedure follows the same as the cold flow's and intended to achieve successful engine ignition. The pressure drops and thrust outputs were measured for comparison against design specification to assess the sealing system's impact on fuel pressurisation and delivery. The pressure drop between the fuel tank and the valve inlet was used to verify theoretical calculations for reliable thrust engine operation.

Method

The cold flow and hot fire tests used identical setups (CAD model shown in Figure 13), with the fully integrated rocket mounted vertically via launch lugs and secured at the feed system's stringer with ratchet straps. The engine was fixed to the TVC's gimbal, with safety rods replacing actuators. The stand was stabilised with 300kg of sandbags. After filling (procedure detailed in Section 0) and checking for leaks, the servo valves were opened to initiate flow to engine.

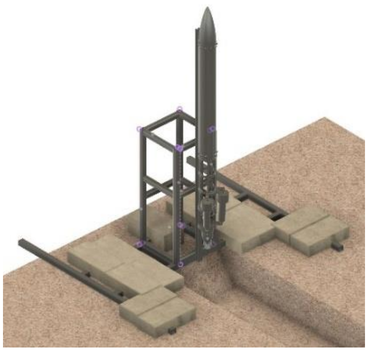

Figure 13 Integrated super-project testing setup

The piping layout is shown in Figure 6. Three pressure transmitters were used,m which enabled real-time monitoring via a Grafana dashboard hosted on a Raspberry Pi connected to a local network. Valve actuation was managed remotely through ICLR's electronic platform, ensuring safe control during ignition and combustion. Two hot fires were performed with varying oxidiser-to-fuel (OF) ratios. In the first, both valves were fully opened ('full-bore'), resulting in a high fuel flow and therefore low OF ratio. In the second, the oxidiser valve remained fully open while the fuel valve was reduced by 40°. This higher OF ratio produced a leaner, hotter combustion regime.

Limitations

A key limitation was the placement of the PTs, which were at a distance from the tank due to bulkhead design. This introduced minor pressure losses to the readings, leading to an underestimation of true tank pressure and making comparisons between the two tanks more difficult. There was also no PTs taking readings at the engine inlet, incurring complications in data analysis. Additionally, PTs are highly sensitive to environmental conditions, so the thermal and vibrational loads during hot fire may have affected data reliability.

Another limitation was the 50Hz sampling rate imposed by ICLR's electronics system, compromising resolution of high-frequency transients during ignition and combustion. Although sufficient for general trends and live monitoring, it may have missed rapid fluctuations. To address this, a separate highspeed logging unit sampling at 500Hz was used. However, an SD card failure meant this higher-resolution data was only available for PT2, limiting analysis of fast dynamics at other points.

The system used ¼” piping throughout to standardise fittings and maintain compatibility with ICLR’s supplied components. As a result, mass flow rates were governed by the servo-valve angle on test day. In the absence of flow meters, flow estimates relied on pressure differentials between PTs, and valve angle adjustment were made through trial and error between hot fires. This limited our ability to test under precisely defined conditions.

The piston’s position during filling proved to be a problematic unknown as it could not be determined until a measured volume of ethanol was added. As a result, the first hot fire was conducted with only ~80% of the intended fuel capacity.

Predicted Measurements

Pressure drop predictions were calculated using the pipe flow energy equation (PFEE) from Equation 1, accounting for all minor losses across fittings and bends.

$$\frac{P_1}{\rho g}+\frac{{U_1}^2}{2g}+z_1=\frac{P_2}{\rho g}+\frac{{U_2}^2}{2g}+z_2+h_L-h_p \tag{1}$$

The pressure of the oxidiser was taken as 50 bar, based on nitrous vapour pressure at 20°C. A conservative estimate of piston friction led to a predicted fuel tank pressure of 49 bar. For the target mass flow rates (calculated from desired thrust and OF ratio), calculated pressure drops from tank to engine inlet were 9.94 bar for fuel and 12.03 bar for oxidiser. Full calculations can be found in Appendix F. Including injector losses of 11 bar (fuel) and 10 bar (oxidiser), the resulting chamber pressure was accepted as a suitable trade-off, balancing pressure delivery and piping integration constraints without compromising engine performance requirements.

The mass flow rate targets for the fuel and oxidiser were 0.16 kg/s and 0.37 kg/s respectively, which were designed around the targeted OF ratio of 2.3. In operation, mass flow rates were controlled by servo valve actuation.

Measured Performance & Discussion

Pressure data from each test was smoothed and plotted in MATLAB. Due to suboptimal PT placement, accurate analysis required several approximations. Analysis of the nitrous lines was especially difficult as only one PT was attached on the dip tube of the lower tank. Without a second PT downstream, mass flow rate estimation was not possible, an oversight in the test instrumentation setup.

Cold flow

Using MATLAB, pressure readings from the three PTs were plotted alongside chamber pressure (Figure 14). The data shows an expected steady decay from a starting pressure of 7.2 bar, lower than in hot fire tests due to nitrogen tank regulation. The average pressure drop across the fuel line (from fuel tank to end of raceway) was approximately 0.5 bar, from which a mass flow rate of 0.053 kg/s was derived as detailed in Appendix G. This is significantly below theoretical predictions, likely due to the indirect measurement method used, lower starting tank pressure and absence of combustion as indicated by the atmospheric chamber conditions (**Figure** 14). A 0.2 bar average pressure difference between the

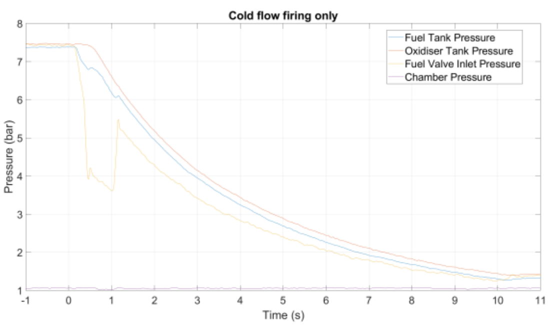

Figure 14 Cold flow pressure data

oxidiser and fuel tanks, shown in the graph, can be attributed to piston friction. The corresponding resistive force can then be calculated to be 205 N.

Hot fire analysis

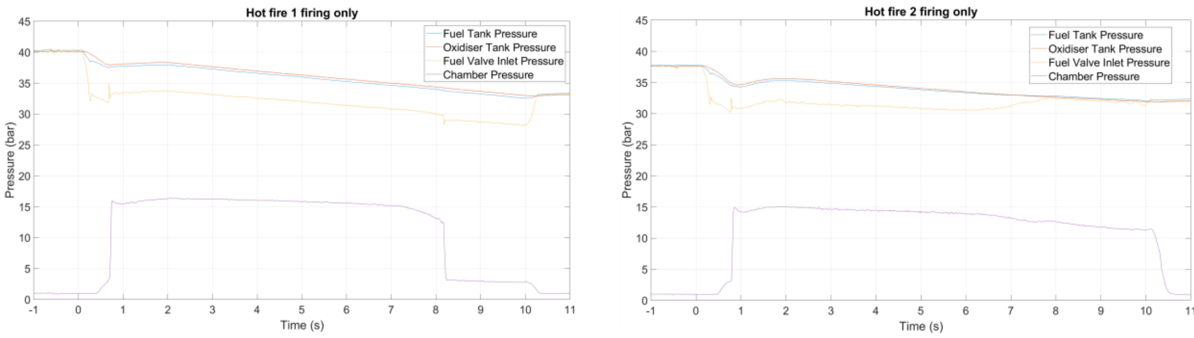

Figure 15 Hot fire pressure data for test 1 (left) and test 2 (right)

Figure 15 (left) shows pressure data collected from the first hot fire test. Tank pressure began at 40 bar, below the 50 bar design value due to lower-than-expected ambient temperature. A small offset between the PTs is visible in the static region and likely results from minor calibration errors in the starting values of the PTs; for the purposes of this analysis and the offset scale relative to other features, this will be treated as negligible. Once the valves opened, a brief transient period was followed by a steady linear pressure decay as mass is consumed. The slight pressure difference between tanks reflects piston friction, while a larger drop between the fuel tank and servo inlet indicates frictional losses around the raceway. The matched decay rates of fuel and oxidiser pressures confirm that the piston moved smoothly throughout.

At around 8 seconds, chamber pressure drops abruptly to a new steady state while feed pressures remain steady, suggesting combustion ended prematurely. This corresponds to a brief fluctuation at the fuel valve inlet, which seems to shut off the combustion before chamber returns to atmospheric pressure when valves are closed at 10 seconds. The cause is uncertain but may be due to momentary fuel starvation, just enough for combustion to end before flow continues for the remainder of the test. The continued pressure decay in both tanks suggests that flow persisted, reinforcing this theory.

During the steady state period of the first hot fire, the average pressure drop measured between the fuel tank and fuel valve inlet was 4.00 bar, as calculated in MATLAB. This value was used in the PFEE to estimate a fuel mass flow rate of 0.124 kg/s, slightly below the design target of 0.160 kg/s (approximately 22% difference). The same flow rate yields a total pressure drop of 7.16 bar, also below the theoretical value, illustrating the correlation between flow rate and pressure losses in system. Only 1 litre of ethanol was filled before it began overflowing from the vent servo, 1.1 litres below tank capacity. This suggests the piston may not have been at the bottom of its motion prior to filling, supporting the idea that the ethanol may have emptied prematurely.

The average pressure difference between the oxidiser and fuel tanks was 0.37 bar, equating to a piston friction force of 375 N. This is an order of magnitude lower than our initial incorrect 3660 N prediction (from a literature-sourced equation (14) detailed in Appendix E), likely due to extrapolating beyond the tested O-ring diameter range. The measured value validates the final 500 N predicted value, from last year's similar design's test.

Hot fire 2 (Figure 15 right), exhibits similar features to the first test but begins at a lower initial pressure of 35 bar, likely due to residual cooling from previous nitrous venting. As in test 1, after a short transient period, the system enters steady-state behaviour. Also similarly, oxidiser and fuel tank pressures decrease in parallel, indicating smooth piston travel without getting stuck.

Unlike the first test, the fuel servo inlet pressure shows oscillations during the steady state, returning to near tank pressure between 7 and 8 seconds. This would suggest a significant drop in fuel flow, however, the chamber pressure remain stable, which indicates sustained combustion for 10 seconds. Additionally, recorded footage shows combustion lasts until the valves are closed. One possible explanation is blockage of the engine's film cooling holes, which may have caused sufficient local residual fuel pooling to sustain combustion without feed. This is consistent with evidence of a minor burn through after the test. Further testing is required to fully characterise this behaviour and determine its cause.

As the behaviour is largely unstable, a shorter window is averaged to find the fuel pressure loss to be 5.64 bar and a corresponding mass flow rate of 0.112 kg/s. This is a lower value than the first test, as is expected from the reduced servo valve angle, which reduced the flow area to achieve a higher OF ratio closer to the design target. Given the uncertainty in the pressure signals, no piston friction calculation was performed for this run's data.

Conclusions and Implications for Design

These tests successfully demonstrated full system integration and produced two hot fires, representing 20% of the target reusability outlined in the PDS. No damage was sustained to the feed system and the piston moved smoothly across both tests, suggesting this subassembly is capable of repeatable operation. Measured fuel mass flow reached 78% of the predicted value and responded predictably to servo valve angle changes. To bring flow closer to the 0.16 kg/s target, a larger internal pipe diameter could be beneficial to reduce frictional losses relative to the flow area. The addition of flow meters on both lines would offer a direct and reliable means of measurement. They could be used to calibrate the servo valve angles during cold flow testing to achieve more accurate mass flow rates for hot firing.

The estimated piston seal friction did not match the expected theoretical prediction of 3660N, proving inaccuracy of the equation when extrapolating to larger O-ring dimensions. A controlled piston sliding test is recommended to explicitly characterise this friction and better understand dynamic sealing behaviour.

To increase the benefit of future hot fire testing, additional pressure transmitters at the engine inlets would provide a basis for better flow rate calculation and a reference point at the feed and engine interface for better monitoring and problem area identification. Further testing is needed to confirm system repeatability and better understand the unexplained transient behaviours observed.

Conclusions & Recommendations for Further Work

A compact, self-pressurising piston tank design was successfully delivered, with a full piping system contained within a uniform silhouette. Ethanol and nitrous oxide were delivered at a pressure and mass flow rate sufficient to support engine burn, which reached the required duration of 10 seconds in test 2. The measured fuel mass flow rate in the first hot fire was 0.124 kg/s, 78% of the target value, followed by 0.112 kg/s in the second (70%), values which could be improved in the future by increasing the internal piping diameters. This subassembly met the dry mass target as specified in the PDS, integrating effectively with flexible piping passing through the gimbal assembly without interference.

Hydrostatic testing confirmed structural integrity, with the tank holding approximately 80 bar for one hour without rupture. Both bulkheads and piston seals performed reliably, with only a minor leak observed from a Swagelok connection. This validated the system for hot fire testing from a safety standpoint. Hot fire results demonstrated smooth piston motion across both test runs. This test also validated the system's reusability, with the feed system having cycled for 20% of the specified target without any observable degradation of the structural components or sealing integrity.

Limitations from sensor placement and logging infrastructure reduced the fidelity of pressure and flow data, impacting post-test data analysis. This highlights the need for greater focus on data acquisition in future testing. To accurately quantify OF ratios, PTs should be included at the injector inlet, and flow meters integrated into the system. These would enable monitoring of propellant flow rates reaching the engine for better calculation of mixture ratios. Additionally, the absence of a PT at the oxidiser line's engine inlet prevents any mass flow calculations for nitrous from being carried out. A bulkhead redesign is also advised to improve accessibility during assembly and create space for mounting PTs directly into the tank for improved measurement accuracy. These upgrades would work together to allow for more detailed combustion analysis, as well as better fault diagnosis and verification of theoretical predictions.

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Appendices

A. Expenditure in detail

Component	Supplier / Manufacturer	Cost
Aluminium Round Bar 6082T6 5/8"	Metal Supermarkets	£22.50
Aluminium Round Bar 6082T6 5" - 260mm	1st Choice Metals	£111.73
Aluminium Round Tube 6082 T6 127x6.3 mm	Aluminium Warehouse	£110.34
Tap sets ((TS) BSPT 1/2 x 14 AND (TS) BSPP 1/4 x 19)	AVON	£62.75
Dip Tube (Seamless 316 Stainless Steel Tube)	Nero	£54.77
Piston (mild steel)	Metal Supermarkets	£150.85
Viton O-rings	Barnwell	£61.61
Stringer - external manufacture	5750 Components	£432.00
EPDM Block	The Metal House	£6.84
Jubilee Clips	Screwfix	£38.68
Torx Bolts	Westfield Fasteners	£30.59
Burst Discs	Wehberg Safety GmbH	£208.41
SPRV	Highpower Systems UK Ltd	£45.00
Washers & nuts	ME Stores	£2.40
Isopropyl Alcohol	Amazon	£7.49
Loctite 577	Amazon	£22.89
O-Ring Lubricant	Simply Bearings	£25.50
Piping system	Swagelok	£599.05

Figure 16 Final expenditure table

B. Tank extension calculations

Calculations were performed for the conditions of 80 bar internal pressure, P, as this is the maximum observed during testing and provides the largest displacement which is easiest to measure from video analysis.

The internal bulkhead area is a circle of diameter 114.3 mm and thus has an area $\mathrm{A_b} = 0.010261\mathrm{m}^3$.

The tank measures 114.3 mm internal diameter and 127 mm external diameter, this gives a wall area of $A_w = 0.002407m^3$. Subtracting the 10 x 11mm bolt holes, the minimum cross-sectional area can be obtained as $\mathrm{A_{w,\,min}} = 0.001772\mathrm{m}^3$, this is used to calculate a range in which we expect the total extension to lie.

Using the following, the maximum and remote wall stresses can be calculated:

$$\sigma_{w,\,max} = \frac{PA_b}{A_{w,\,min}} = \frac{80 \times 10^5 \times 0.010261}{0.001772} = 46.325 \text{ MPa}$$

$$\sigma_w = \frac{PA_b}{A_w} = \frac{80 \times 10^5 \times 0.010261}{0.002407} = 34.103 \text{ MPa}$$

Treating the holes as a stress concentrating feature with a factor of 3, the remote stress can be used to determine the hole wall stress:

$$\sigma_h = SCF \times \sigma_w = 3 \times 34.103 = 102.309\ MPa$$

Now using the Youngs modulus of aluminium 6082T6 which is 71 GPa, the strain and thus extension can be determined as

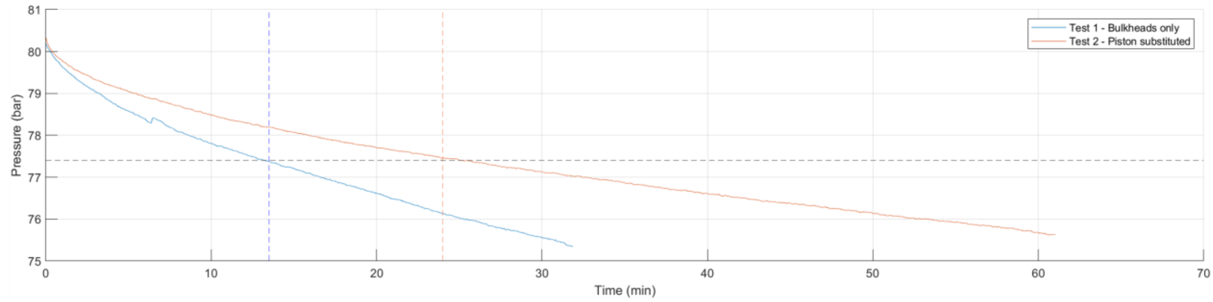
$$\Delta \mathrm{L} = \frac{\sigma \mathrm{L}}{E}$$

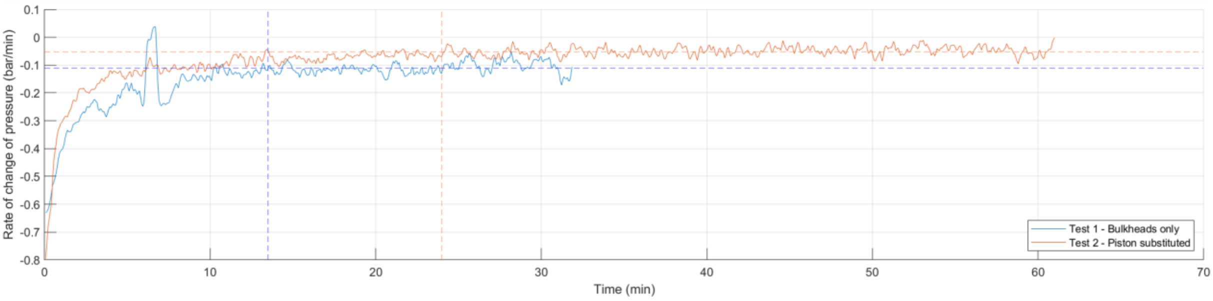
For the tank, the initial length L is 959 mm and for the holes, the initial length is 11 mm. Substituting the parameters into the above equation it can be found that the full tank extension is 0.62 mm when using the full area and 0.46 mm when using the minimum cross-sectional area. This suggests the real extension of the tank should lie in the range 0.46 mm - 0.62 mm.

Similarly, the strain expected at the hole wall works out to be 0.001441 (an extension of 0.02 mm) which can be directly compared to the results of the DIC analysis.

C. Constant rate of change inspection

By averaging the second half of the rate of pressure change data from both tests and drawing a horizontal line across at these values, lines can be drawn up at 13.5 min for test 1 and 24 min for test 2 where the leak rates start to become constant to their respective averaged values. Upon inspection of the pressures at which this point occurs, an interesting relationship can be observed as both leaks appear to become constant when the tank pressure is reaches approximately 77.4 bar.





D. DIC analysis

ZEISS INSPECT Correlate Screenshots

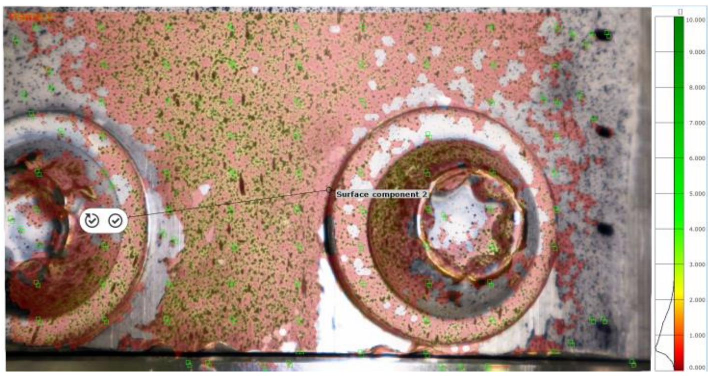


Figure 17 ZEISS INSPECT pattern quality analysis

The pattern quality, shown in Figure 17, scores a 1/10 which is a very poor base for analysis. This is due to the coarse pattern resulting from the near empty spray cans and the damage sustained from reassembly which can be clearly seen around the bolt locations.

Figure 18 shows the results for vertical displacement (left) and strain (right) experienced under full 80 bar loading. The average displacement from the points marked on the tank is 0.921 mm, the bolts show greater displacement of about 2 mm as the bulkheads settle and push out of the tank under pressure. The local strain output has a negligible average thus provides no useful data.

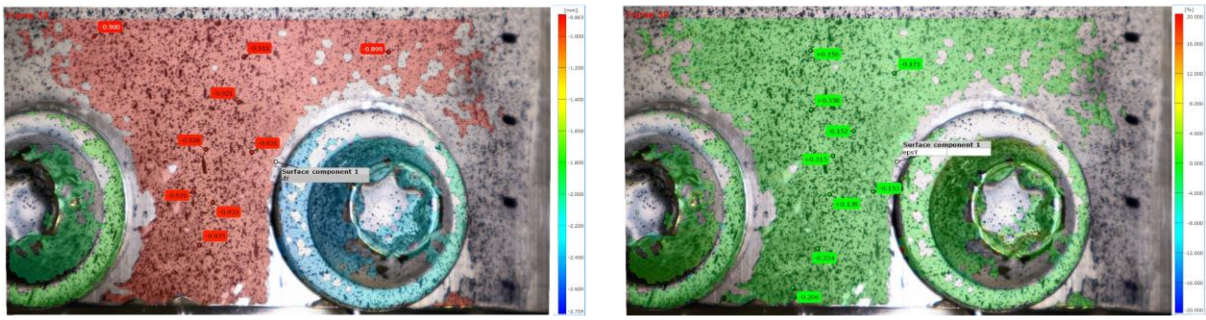


Figure 18 ZEISS INSPECT results for vertical displacement (left) and local strain (right)

GIMP image manipulation pixel-analysis

Shown below are the full results of the GIMP image manipulation software analysis which uses the four 10 mm spaced markings visible to the right side of Figure 18 (these marks are referenced as numbered top to bottom). The analysis is carried out by counting pixel distances between features across different frames of the recording to obtain information about the loading conditions.

The average distance between the four markings in frame 2, acts as the reference for the camera scale. This averages to 581 pixels and as the markings are known to have 10mm spacing under no load, it can be concluded that there are 58.1 pixels per mm across the image.

1. Distance between markings

Point No	Frame 2 spacing to the following point	Frame 59 spacing to the following point
1-2	578	581
2-3	578	583
3-4	587	582
Average	581	582

2. Motion of individual markings

Point No	Frame 2 – 59 marking movement
1	50
2	49
3	54
4	50
Average	50.75

3. Bolt movements

Point No	Frame 2 – 59 movement
1	62
2	77
Average	70

The local extension, determined by taking the difference between the no load (frame 2) and full 80 bar load (frame 59) marking spacings in table 1, is 1 pixel which equates to a 0.0172 mm extension.

The global extension of the tank is obtained from the motion of the individual markings between the 2nd and 59th frames which are given in table 2. Here the average motion is 50.75 pixels which is an extension of 0.873 mm.

Finally, the motion of the bolts can be determined relative to the tank to observe how much the bulkheads push out under pressure. This is table 3 above and has an average of 70 pixels equating to a 1.2 mm movement. It is possible some of the tank displacement observed in the video frame analysis comes from a similar settling between the tank and bottom bulkhead. This motion has not been accounted for in the full tank strain as the behaviour at this end is not visible in the video data. As this effect is likely to affect the results of the full tank strain analysis, it should be addressed in further testing by either a duplicate camera setup at the other end or through use of a bonded strain gauge.

E. Piston pressure drop calculations

The pressure drop resulting from the friction between the O-ring seals and the tank was determined. The friction force exerted by the seals was computed using the following equation (14)

$$F = 2 \cdot N \cdot \pi \cdot D \cdot r \cdot E \cdot \left(1 - \frac{D-d}{4r}\right)\sqrt{1 - \frac{(D-d)^2}{16r^2}}$$

Symbol	Meaning	Value for the piston
N	Number of seals at interface	2
D	Outer diameter of sealing interface	114.3 x 10^{-3} m
d	Inner diameter of sealing interface	105.6 x 10^{-3} m
r	Cross sectional radius of the O-ring	2 x 10^{-3} m
E	Elastic modulus of the material	For Viton 9 x 10^{6} Pa

Using these values in the equation gives 3660 N as the predicted frictional force at the piston sealing interface. It is again noted that this value is extrapolated from literature and did not agree with real measurements taken in the previous years project.

F. Pipe Flow Energy Equation

The pipe flow energy equation was used to produce estimations for pressure drops and mass flow rates, the general method for which is described below.

$$\frac{P_1}{\rho g} + \frac{{U_1}^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{{U_2}^2}{2g} + z_2 + h_L - h_p$$

P_1 and P_2 refer to the upstream and downstream pressures of the propellants. ρ and g refer to the density of the propellants and gravity respectively. U_1 and U_2 refer to the upstream and downstream velocities, however as they are assumed to be the same, their terms are cancelled in the equation. z_1 and z_2 refer to the elevation of the propellants upstream and downstream of the system. Finally, h_L refers to the total head loss in the propellants, and h_p refers to pump loss, but due to the absence of pumps in the system this term is neglected.

To calculate the velocity of the propellants, their predicted mass flow rates, $\dot{m}$, are used alongside their density and the cross-sectional area of the piping, A, as shown in the equation below.

$$U = \frac{\dot{m}}{\rho \times A}$$

The total head loss is sum of the minor losses, h_l, and major losses, h_f.

$$h_L = h_f + h_l$$

The equation below finds the minor losses, which requires K, the sum of all the minor loss coefficients of the components in the system.

$$h_l = K\frac{U^2}{2 \times g}$$

The equation below finds the major losses

$$h_f = \frac{f \times L \times U^2}{2 \times g \times D}$$

L refers to the total length of the piping system and f is found via the Haaland equation

$$\frac{1}{\sqrt{f}} = 1.8log_{10}\left[\left(\frac{\frac{\varepsilon}{D}}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right]$$

Where ε refers to the absolute roughness of the piping.

The Haaland equation requires the Reynolds number of the propellants, Re, which is calculated using

$$Re = \frac{\rho \times U \times D}{\mu}$$

Where D refers to the diameter of the piping, and μ refers to the dynamic viscosity of the propellants.

All required values needed to perform the calculations above included below.

Pipe properties	Ethanol	Nitrous
Pipe ID [mm] *(316L pipe/ flexible hose)*	4.52/4.8	4.52/4.8
Length [mm] *(316L pipe/ flexible hose)*	1735/304.8	786.6/457.2
Minor losses coefficient	2.95	1
Absolute pipe roughness [mm] *(316L pipe/ flexible hose)*	0.015/0.0015	0.015/0.0015

Fluid properties	Ethanol	Nitrous
Tank pressure [bar]	49	50
Mass flow rate [kg/s]	0.16	0.37
Density [kg/m^3]	789	786.6
Dynamic viscosity [Pa s]	0.0011841	0.0000147

G. Mass flow rate and pressure drop calculations using measured data

Using MATLAB, pressure readings from the three feed system pressure transmitters were plotted alongside the engine chamber pressure. From the plots, a steady state section was selected and the average pressure difference between the fuel tank pressure readings and the fuel valve inlet pressure readings was calculated. The mass flow rate was then estimated based on this pressure difference via the pipe flow energy equation described in Appendix F. The length of the piping system and the minor loss coefficient were adjusted to reflect only the distance between the two pressure transmitters that produced the readings. Using these properties, the pipe flow energy equation was iterated until a mass flow rate was found that agrees with the observed pressure difference.

This method was applied for both the cold flow and hot fires, for which the adjusted properties are shown below.

Pipe properties	Values
Pipe ID [mm] *(316L pipe/ flexible hose)*	4.52/4.8
Length [mm]	1292
Minor losses coefficient	2.3
Absolute pipe roughness [mm]	0.015

Fluid properties	Distilled water	Ethanol *(Hot fire 1/ Hot fire 2)*
Tank pressure [bar]	7.38	37.9/35.3
Average pressure drop from PT1 to PT2 [bar]	0.5	4.00/3.31
Density [kg/m^3]	998.2	789
Dynamic viscosity [Pa s]	0.0010005	0.0011841

The estimated mass flow rates were 0.053 kg/s in the cold flow, 0.124 kg/s in the first hot fire, and 0.112 kg/s in the second hot fire.

The estimated mass flow rates for the first and second hot fires were then used to estimate a total pressure drop across the piping system. This was done by using the same method and properties described in Appendix F but replacing the target mass flow rate of 0.16 kg/s for the fuel.

The estimated pressure drops were 7.16 bar in the first hot fire, and 5.64 bar in the second hot fire.